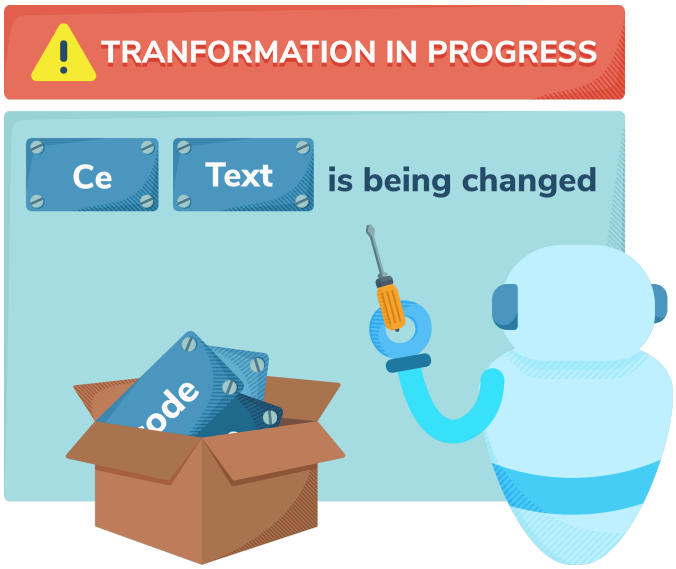
Transformation

Explore how to leverage OpenAI's completions endpoint to perform text transformations including translating between languages, summarizing lengthy text, and converting text into emojis. Gain practical skills in designing prompts and adjusting parameters to create effective NLP solutions using Python.

Overview

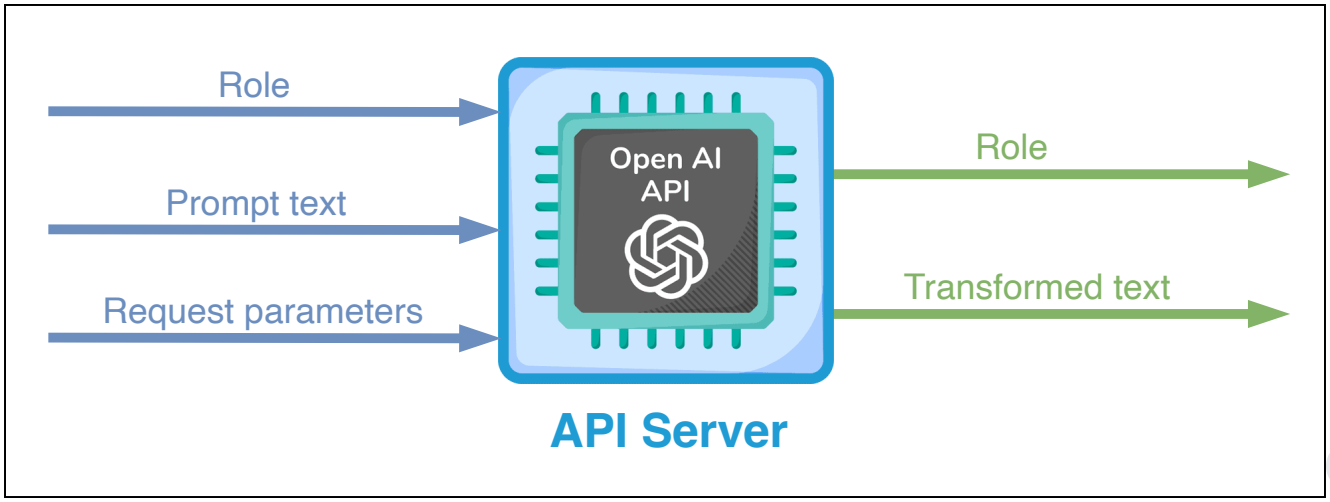
The chat completions endpoint has the ability to translate from one language to another language, summarize the text, and convert the text to emojis or vice versa.



Transformation

Ask

We input the prompt text and request parameters into the OpenAI API's model to transform the text.



Completions: text transformation

Translation

The OpenAI API can translate from one language to another language.

Let's see how to provide the prompt to translate from English to other languages.

```
Translate this into 1. French, 2. Spanish and 3. Japanese.

We provide an online learning platform made by developers, created for develo
```

Example

Let's try the above example in the code widget below:

Please provide values for the following:

SECRET_KEY Not Specified...

Ask

Edit Import Values from JSON

main.py Python

```python
#Translating from english to other languages

from openai import OpenAI
client = OpenAI(api_key="{{SECRET_KEY}}")

response = client.chat.completions.create(
  model="gpt-3.5-turbo",
  response_format={ "type": "json_object" },
  messages=[
    {"role": "system", "content": "Translate this into 1. French, 2. Span
    {"role": "user", "content": "We provide an online learning platform m
  ],
  temperature=0
```

Run

Translating from english to other languages

In the code above, `temperature` is set to `0` because we don't need randomness in the translation, and the value of `max_tokens` is set to `100` because we know the approximate length of output tokens. All other parameter values are the default values.

Similarly, we can design a prompt to translate it back into English:

```
Translate the following into the English language:

1. Nous fournissons une plateforme d'apprentissage en ligne créée par des dé
2. Proporcionamos una plataforma de aprendizaje en línea creada por desarrol
3.      のために
```

Ask

Summarization

The OpenAI API models have been trained enough to efficiently summarize lengthy texts. They can not only learn the context of the provided text but also have the ability to rewrite it in easy language.

Let's see how to provide the prompt design to summarize a lengthy text:

```
Summarize the following text:

<Text to summarize>
```

Example

Let's try an example in the code widget below:

Please provide values for the following:

SECRET_KEY Not Specified...

Edit Import Values from JSON

main.py Python

```python
# Summarizing the text

from openai import OpenAI
client = OpenAI(api_key="{{SECRET_KEY}}")

response = client.chat.completions.create(
  model="gpt-3.5-turbo",
  response_format={ "type": "json_object" },
  messages=[
    {"role": "system", "content": "Summarize the given text for a second-
    {"role": "user", "content": ("Jupiter is the fifth planet from the "+
                "Sun and the largest in the Solar System. It is a g
                "one thousandth that of the Sun, but two and a hal
```

Ask

Run

Summarizing the text

In the code above, we use `temperature` with a higher value of `0.9` to generate a more abstract summary of the text. The parameters `frequency_penalty` and `presence_penalty` are set to `1` because the new tokens will have a greater chance to appear, and tokens will be repeated fewer times. We set the output text length to `64`.

Conversion

OpenAI has been trained on emojis as well and can convert text into emojis that correspond to the meaning of the text.

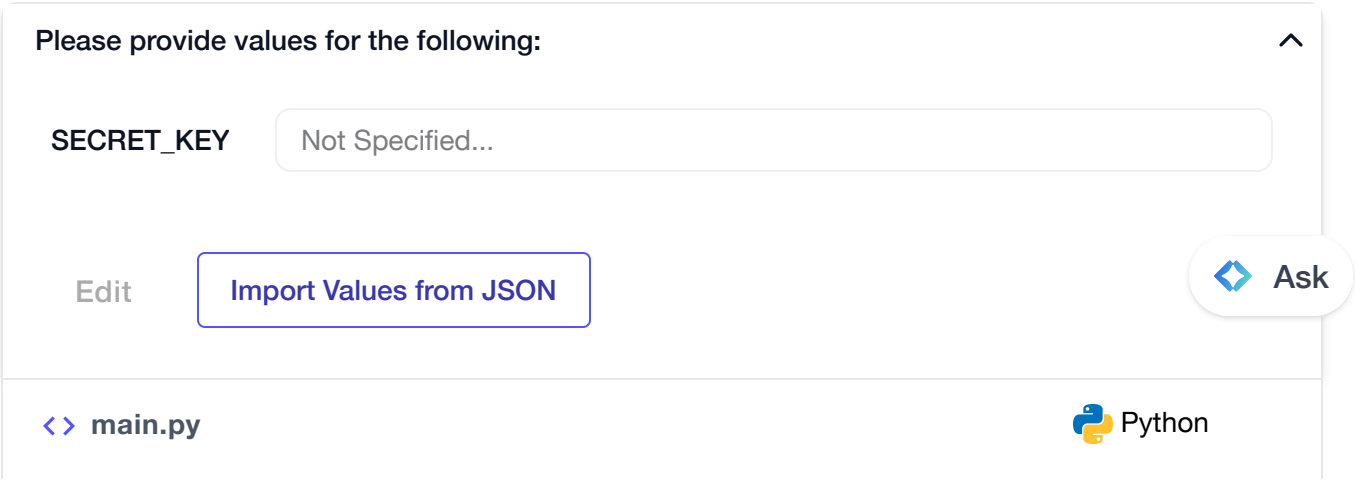
Let's see how to provide the prompt to convert movie names to emojis. In the following prompt, we have provided four movie names in the first three examples with their corresponding emojis, and the last one is the movie for which we require an emoji.

```
Convert movie titles into emojis.

Back to the Future: 👨👴🚗🕒
Batman: 🤵🦇
Transformers: 🚗🤖
Antman:
```

Example

Let's try the above example in the code widget below:



```
#Conversion of text to emojis

from openai import OpenAI
client = OpenAI(api_key="{{SECRET_KEY}}")

response = client.chat.completions.create(
  model="gpt-3.5-turbo",
  response_format={ "type": "json_object" },
  messages=[
    {"role": "system", "content": "Convert movie titles into emojis. Prov
    {"role": "user", "content": ("Back to the Future: 👨👦🚗🕒\n" +
                "Batman: 🤵🦇\n" +
```

Run

Conversion of text to emojis

We set the `temperature` value to `0.9` to add creative generation. You can experiment by changing the `temperature` value and `engine` and find an appropriate value for these parameters.

Practical use case

Some practical use cases of OpenAI's transformation capabilities include:

- **Cross-language content summarization**: A news aggregator platform collects articles in multiple languages worldwide. OpenAI wants to provide concise summaries of articles to users, regardless of the original language. The platform utilizes the OpenAI chat completions endpoint to summarize articles in their original language. Then, it translates the summaries into the user's preferred language, enabling users to grasp the main points of articles written in diverse languages.
- **Emoji-based language communication**: A social media platform air enhance user engagement by allowing users to communicate using emojis instead of text. Users input text messages, and the platform employs the

Ask

OpenAI chat completions endpoint to convert the text to emojis. The chatbot translates the text into a sequence of relevant emojis, enabling users to

Ask